

QUESTION-10

Stacked Using Array

main.c	Output
<pre>1 #include <stdio.h> 2 #define MAX 5 3 int stack[MAX]; 4 int top = -1; 5 void push(int value) { 6 if (top == MAX - 1) { 7 printf("Stack Overflow\n"); 8 return; 9 } 10 top++; 11 stack[top] = value; 12 printf("%d pushed into stack\n", value); 13 } 14 void pop() { 15 if (top == -1) { 16 printf("Stack Underflow\n"); 17 return; 18 } 19 printf("Deleted Element: %d\n", stack[top]); 20 top--; 21 } 22 void peek() { 23 if (top == -1)</pre>	<pre>10 pushed into stack 20 pushed into stack 30 pushed into stack Stack Elements: 30 20 10 Top Element: 30 Deleted Element: 30 Stack Elements: 20 10 === Code Execution Successful ===</pre>

main.c	Output
<pre>11 stack[top] = value;open screenshot 12 13 printf("%d pushed into stack\n", value); 14 } 15 void pop() { 16 if (top == -1) { 17 printf("Stack Underflow\n"); 18 return; 19 } 20 printf("Deleted Element: %d\n", stack[top]); 21 top--; 22 } 23 void peek() { 24 if (top == -1) 25 printf("Stack is Empty\n"); 26 else 27 printf("Top Element: %d\n", stack[top]); 28 } 29 void display() { 30 int i; 31 if (top == -1) { 32 printf("Stack is Empty\n");</pre>	<pre>10 pushed into stack 20 pushed into stack 30 pushed into stack Stack Elements: 30 20 10 Top Element: 30 Deleted Element: 30 Stack Elements: 20 10 === Code Execution Successful ===</pre>

main.c	Output
<pre>27 printf("Top Element: %d\n", stack[top]); 28 } 29 void display() { 30 int i; 31 if (top == -1) { 32 printf("Stack is Empty\n"); 33 return; 34 } 35 printf("Stack Elements:\n"); 36 for (i = top; i >= 0; i--) { 37 printf("%d\n", stack[i]); 38 } 39 } 40 int main() { 41 push(10); 42 push(20); 43 push(30); 44 display(); 45 peek(); 46 pop(); 47 display(); 48 return 0; 49 }</pre>	<pre>10 pushed into stack 20 pushed into stack 30 pushed into stack Stack Elements: 30 20 10 Top Element: 30 Deleted Element: 30 Stack Elements: 20 10 === Code Execution Successful ===</pre>